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Inventor(s)	Frye; Kevin D.

Wildlife Guard for Cutout Fuse Holder

Abstract

A wildlife guard for protecting an electrical distribution cutout fuse holder from electrical shorts caused by wildlife is disclosed. The guard includes a first body piece pivotally connected to a second body piece, thereby creating a cover having an open bottom and an open front for receiving the cutout fuse holder and the insulator. A spring biases the cover to a closed position. The wildlife guard also includes a releasable trigger including a first trigger arm connected to a second trigger arm. The releasable trigger operates to hold the cover in the open position until contacted by the cutout fuse holder, which converts the cover to the closed position. The wildlife guard also includes spikes attached to the inner surfaces of the body pieces that impede wildlife from crawling under the cover.

Inventors:	Frye; Kevin D. (Star City, AR)
Applicant:	Central Moloney, Inc. (White Hall, AR)
Family ID:	1000007819127
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] Not applicable

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not applicable

BACKGROUND OF THE INVENTION

[0003] Squirrels, birds, and the like often climb onto or perch on transformers, power lines, insulators, and other types of electrical distribution equipment. If the distance between an energized terminal and a ground or another terminal is too short, an animal may come into contact with both at the same time. This will typically cause an electrical short and kill the animal. It may also damage equipment and cause power outages. Wildlife guards are typically used as a barrier to maintain adequate space between the animal and the energized part to decrease the chances of an animal causing an electrical short. But installing wildlife guards can be time-consuming and dangerous because the installer often must be lifted high above the ground to reach the electrical distribution equipment. The entire line circuit must also be de-energized to perform this operation. And installing wildlife guards from the ground can be difficult. It would therefore be desirable to develop a wildlife guard for a cutout fuse holder and accompanying electrical distribution equipment that is easily installable from the ground without the need to de-energize the line circuit.

BRIEF SUMMARY OF THE INVENTION

[0004] The present invention is directed to a wildlife guard for protecting a cutout fuse holder that is attached to an insulator used in electrical distribution from shorts caused by wildlife. In one preferred embodiment, the wildlife guard includes a first body piece pivotally connected to a second body piece, thereby creating a cover having an open bottom and an open front for receiving the cutout fuse holder and the insulator. A spring biases the cover to a closed position. The wildlife guard also includes a releasable trigger including a first trigger arm connected to a second trigger arm. The first trigger arm is attached to the inner surface of the first body piece and the second trigger arm is attached to the inner surface of the second body piece. The releasable trigger is configured to hold the cover in the open position until contacted by the cutout fuse holder, which converts the cover to the closed position. The wildlife guard also includes spikes attached to the inner surfaces of the body pieces that impede wildlife from crawling under the cover.

[0005] In another preferred embodiment, the present invention is directed to a method of installing a wildlife guard on a cutout fuse holder attached to an insulator comprising the steps of: (1) providing a wildlife guard of the present invention; (2) converting the cover of the wildlife guard from the closed position to the open position by pulling the first body piece away from the second body piece to activate the trigger; and (3) pushing the wildlife guard on top of the cutout fuse holder and the insulator until the cutout fuse holder contacts the trigger, thereby causing the spring to convert the cover from the open position to the closed position.

[0006] These and other features, objects and advantages of the present invention will become better understood from a consideration of the following detailed description of the preferred embodiments and appended claims in conjunction with the drawings as described following:

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a perspective view of an insulator attached to a cutout fuse holder with a fuse in the closed position.

[0008] FIG. 2 is a perspective view of an insulator attached to a cutout fuse holder with a fuse in the open position.

[0009] FIG. 3 is a perspective view of a cutout fuse holder with an attached parallel groove connector.

[0010] FIG. 4 is a top perspective view of the wildlife guard of the present invention.

[0011] FIG. 5 is a perspective view of the wildlife guard of the present invention covering the top of an insulator attached to a cutout fuse holder.

[0012] FIG. 6 is a side view of an insulator attached to a cutout fuse holder with one body piece of the wildlife guard of the present invention covering the top of the insulator and the cutout fuse holder.

[0013] FIG. 7 is a bottom perspective view of the wildlife guard of the present invention in the closed position.

[0014] FIG. 8 is a bottom perspective view of the wildlife guard of the present invention in the open position.

DETAILED DESCRIPTION OF THE INVENTION

[0015] With reference to FIGS. 1-8, the preferred embodiments of the present invention may be described. The present invention is a wildlife guard **10** for protecting a cutout fuse holder **12** that is attached to an insulator **14** used in electrical distribution from shorts caused by wildlife. As shown in FIGS. 1-2 and as well understood by a person of ordinary skill in the art, the insulator **14** is mounted to a utility pole **16**. The cutout fuse holder (or cutout) **12** is attached to the top and bottom of the insulator **14** by a top arm assembly **18** and a bottom arm assembly **20**. The fuse **22**, which is a fuse tube, is inserted between the top arm assembly **18** and the bottom arm assembly **20**. When the fuse tube **22** is contacting the top arm assembly **18** and the bottom arm assembly **20**, the fuse **22** is in the closed position, as shown in FIG. 1. When the fuse tube **22** is only contacting the bottom arm assembly **20**, the fuse **22** is in the open position, as shown in FIG. 2. The top arm assembly **18** includes attachment hooks **24** for guiding the fuse during insertion and closing, an upper contact **26** that is contacted by the end of the fuse tube, a spring **28** positioned above the upper contact **26** to ensure a tight fit of the tube fuse between the top arm and bottom arm assemblies, and a top plate **30**. A parallel groove connector **32** is fastened to the top plate **30** of the top arm assembly **18**. The parallel groove connector **32** is a wire connector. A wire is connected to the parallel groove connector **32** and the overhead power line. It is desirable to install the wildlife guard **10** over the top of the insulator **14**, parallel groove connector **32** and top plate **30** of the top arm assembly **18** to maximize the insulating and protective value of the wildlife guard **10**. The wildlife guard **10** has features that allow it to be easily aligned and installed on the top of the insulator **14** and the top plate **30** to protect the parallel groove connector **32** and the top plate **30** from being contacted by wildlife. Wildlife simultaneously contacting the top plate **30** or parallel groove connector **32** and the nearby ground node will cause the fuse to blow

[0016] As shown in FIG. 4, the wildlife guard **10** comprises two body pieces **34** of insulating material. Each body section **34** includes a back section **36**, a first top section **38**, a second top section **40**, and a side section **42**. The back section **34** is curved towards the side section **42**. The top of the back section **34** also is curved towards the first top section **38**. The first top section **38** is positioned between the back section **36** and the second top section **40**. The first top section **38** is curved towards the side section **42**. The second top section **40** is also curved towards the side section **42**. In addition, the second top section **40** slopes downwardly towards the front of the wildlife guard. A free edge of the second top section **40** is the front edge of the wildlife guard. The front and bottom of the wildlife guard **10** are open for receiving portions of the insulator **14** and cutout fuse holder **12**.

[0017] The body pieces **34** are pivotally connected together along their second top sections **40**, as shown in FIGS. 4 and 8. The pivotal attachment may be by hinges **44**, although the present invention is not limited to pivotal connection using hinges. Other forms of pivotal connection known in the art would also be acceptable.

[0018] When the two body pieces **34** are in the closed position as shown in FIGS. 4-5, the two body pieces **34** form an irregular shaped cover. As shown in FIGS. 4, each first top section **38** has a concave edge **46**. The concave edge **46** is provided with a plurality of flexible fingers **48**. The

flexible fingers **48** from the opposing body pieces **34** face each other when the guard is in the closed position. The flexible fingers **34** allow a conductor (such as copper wires) to pass through from the interior to the exterior of the guard **10** when the two body pieces **34** of the guard are engaged.

[0019] The guard **10** is biased toward a closed position by a spring **58**. The biasing spring **58** is desirably in the form of a semi-circular loop of resilient material that is connected to the side sections **42** of both body pieces **34**. The spring **58** extends across the second top sections of both body pieces **34**. The biasing spring **58** has short extensions at right angles to fit into slots **50** in the side sections **42**.

[0020] The wildlife guard includes two positioning plates **52** attached to the inner surface of the wildlife guard. The positioning plates **52** are preferably L-shaped plates. A positioning plate **52** extends inwardly from each of the body pieces **34**. In one embodiment, the positioning plates **52** extend from body pieces **34** at an approximately 90 degree angle. The positioning plates **52** serve as stops to prevent the top plate **30** of the cutout fuse holder from extending farther into the wildlife guard **10**. The positioning plates **52** also serve as attachment points for a releasable trigger **54** of the wildlife guard. The inner surface of the body pieces **34** also include spikes **74**. When the body pieces **34** are closed, the spikes of one body piece **34** face the spikes of the other body piece **34**. The spikes occupy otherwise empty space under the wildlife guard **10** when it is positioned on the cutout fuse holder **12** and block wildlife from crawling under the wildlife guard **10**. The spikes also aid in properly positioning the wildlife guard **10** on the cutout fuse holder **12**. If the spikes contact the cutout fuse holder **12**, insulator **14**, or parallel groove connector **32** during installation of the wildlife guard **10**, the wildlife guard **10** is deflected into the proper position on the cutout fuse holder **12**, insulator **14**, and parallel groove connector **32**.

[0021] The releasable trigger **54** is used to hold the two body pieces **34** in an open position until the top plate **30** or parallel groove connector **32** displaces the trigger **54** and allows the biasing spring **58** to close the two pieces **34**. The releasable trigger **54** comprises two trigger arms **56** pivotally connected at their inner ends. The outer ends **64** of the trigger arms **56** are pivotally connected to the positioning plates **52** in the interior of the guard. When the two body pieces **34** are opened against the biasing force of the spring **58** and the trigger arms **56** are in alignment as shown in FIG. **8**, a quasi-stable configuration of the trigger arms **56** results and the two body pieces **34** are held apart against the closing force provided by the biasing spring **58**. This activation (or setting) of the trigger can be accomplished by pulling the two body pieces apart from one another. When the inner pivot point is displaced slightly inwardly by contact with the top plate **30** or the parallel groove connector **32**, the trigger arms **56** are no longer in a stable configuration and continue to pivot inwardly under the force of the biasing spring **58**, thereby allowing the two body pieces **34** to close together as shown in FIG. **7**. A stop **62** prevents the trigger arms **56** from being displaced outwardly from the quasi-stable configuration. Thus, the trigger arms **56** can only be displaced inwardly.

[0022] A handle **66** extends downwardly from the bottom of the side sections **42** of the body pieces **34**. Each handle **66** has an opening **68** with a plurality of inner grooves for engaging and manipulating the guard **10** with a widely-used type of manipulator tool known as shotgun stick. The present invention is not limited to the preferred embodiment of the handle **66** as described above. As used herein, the term "handle" refers to any appendage by which a manipulator tool may grasp the guard **10**. To install the guard **10**, a worker using a shotgun stick pushes the open guard **10** against the top of the top plate **30** of the cutout fuse holder **12**. The trigger **54** is released as described above and the biasing spring **58** snaps the two body pieces **34** into engagement in a closed position to form a cover around the top of the insulator **14**, cutout fuse holder **12**, and parallel groove connector **32**, as shown in FIGS. **5-6**. The spikes **74** fit closely around and underneath the parallel groove connector **32** and the top of the insulator **14** when the two body pieces **34** are in the closed position.

[0023] As noted above, the guard **10** may be installed from the ground with a hotstick. The

structure of a hotstick is described and illustrated in U.S. Pat. No. 6,995,313, which is incorporated by reference herein. The hotstick is designed to be fitted with a variety of tools since the end of the hotstick has a radially-arranged series of wedge-shaped protrusions which interlock with a similar arrangement of wedge-shaped protrusions on an adapter. The releasable adapter engages the handle **66** by sliding the handle **66** into a complementary recess in the adapter. The adapter is provided with an arrangement of radially-arranged wedge-shaped protrusions to complement the arrangement on the hotstick. The adapter is fastened to the end of the hotstick by a wingnut as is conventionally known at any suitable angle to the hotstick as determined by the operator. Once the wildlife guard **10** is installed on the cutout fuse holder **12**, the adapter is easily disengaged from the handle **66** by sliding the adapter off the handle **66**.

[0024] The present invention has been described with reference to certain preferred and alternative embodiments that are intended to be exemplary only and not limiting to the full scope of the present invention as set forth in the appended claims.

Claims

1. A wildlife guard for a cutout fuse holder attached to an insulator, said wildlife guard comprising: a first body piece having an inner surface and an outer surface; a second body piece having an inner surface and an outer surface, wherein said first body piece is pivotally connected to said second body piece, thereby creating a cover comprising an open bottom and an open front for receiving said cutout fuse holder and said insulator, wherein said cover comprises an open position and a closed position; a spring biasing said cover to a closed position, wherein said spring is connected to said outer surface of said first body piece and said outer surface of said second body piece; a releasable trigger comprising a first trigger arm connected to a second trigger arm, wherein said first trigger arm is attached to said inner surface of said first body piece and said second trigger arm is attached to said inner surface of said second body piece, wherein said releasable trigger is configured to hold said cover in said open position until contacted by said cutout fuse holder; and a plurality of spikes attached to said inner surface of said first body piece and said inner surface of said second body piece, wherein said plurality of spikes are configured to impede wildlife from crawling under said cover.
2. The wildlife guard of claim 1, wherein a first handle extends from a bottom of said first body piece and a second handle extends from a bottom of said second body piece.
3. The wildlife guard of claim 1, wherein said spring extends across a top of said cover.
4. The wildlife guard of claim 1, wherein said first body piece comprises a first set of flexible fingers and said second body piece comprises a second set of flexible fingers, wherein said first set of flexible fingers face said second set of flexible fingers, wherein said first set of flexible fingers and said second set of flexible fingers are configured such that a wire attached to said cutout fuse holder can pass in between some of said first set of flexible fingers and said second set of flexible fingers.
5. The wildlife guard of claim 1, further comprising a first positioning plate connected to said inner surface of said first body piece and a second positioning plate connected to said inner surface of said second body piece, wherein said first trigger arm is connected to said first positioning plate and said second trigger arm is connected to said second positioning plate.
6. The wildlife guard of claim 1, further comprising a plurality of hinges.
7. A wildlife guard for a cutout fuse holder attached to an insulator, said wildlife guard comprising: a first body piece having an inner surface and an outer surface, wherein said first body piece comprises a back section, a first top section, a second top section, and a side section; a second body piece having an inner surface and an outer surface, wherein said second body piece comprises a back section, a first top section, a second top section, and a side section; a plurality of hinges attached pivotally connecting said second top section of said first body piece and said second top

section of said second body piece, thereby creating a cover comprising an open bottom and an open front for receiving said cutout fuse holder and said insulator, wherein said cover comprises an open position and a closed position; a spring biasing said cover to a closed position, wherein said spring is connected to said outer surface of said first body piece and said outer surface of said second body piece; and a releasable trigger comprising a first trigger arm connected to a second trigger arm, wherein said first trigger arm is attached to said inner surface of said first body piece and said second trigger arm is attached to said inner surface of said second body piece, wherein said releasable trigger is configured to convert said cover from said open position to said closed position.

8. The wildlife guard of claim 7, further comprising a plurality of spikes attached to said inner surface of said first body piece and said inner surface of said second body piece to impede wildlife from crawling under said cover.

9. The wildlife guard of claim 7, wherein a first handle extends from a bottom of said first body piece and a second handle extends from a bottom of said second body piece.

10. The wildlife guard of claim 7, wherein said spring extends across a top of said cover.

11. The wildlife guard of claim 7, wherein said first body piece comprises a first set of flexible fingers and said second body piece comprises a second set of flexible fingers, wherein said first set of flexible fingers face said second set of flexible fingers, wherein said first set of flexible fingers and said second set of flexible fingers are configured such that a wire attached to said cutout fuse holder can pass in between some of said first set of flexible fingers and said second set of flexible fingers.

12. The wildlife guard of claim 7, further comprising a first positioning plate connected to said inner surface of said first body piece and a second positioning plate connected to said inner surface of said second body piece, wherein said first trigger arm is connected to said first positioning plate and said second trigger arm is connected to said second positioning plate.

13. A method of installing a wildlife guard on a cutout fuse holder attached to an insulator comprising the steps of: providing a wildlife guard comprising: a first body piece having an inner surface and an outer surface; a second body piece having an inner surface and an outer surface, wherein said first body piece is pivotally connected to said second body piece, thereby creating a cover comprising an open bottom and an open front for receiving said cutout fuse holder and said insulator, wherein said cover comprises an open position and a closed position; a spring biasing said cover to a closed position, wherein said spring is connected to said outer surface of said first body piece and said outer surface of said second body piece; and a releasable trigger comprising a first trigger arm connected to a second trigger arm, wherein said first trigger arm is attached to said inner surface of said first body piece and said second trigger arm is attached to said inner surface of said second body piece. converting said cover from said closed position to said open position by pulling said first body piece away from said second body piece to activate said trigger; and pushing said wildlife guard on top of said cutout fuse holder and said insulator until said cutout fuse holder contacts said trigger, thereby causing said spring to convert said cover from said open position to said closed position.
